

AI & Machine Learning Track

From Data to Intelligent Applications

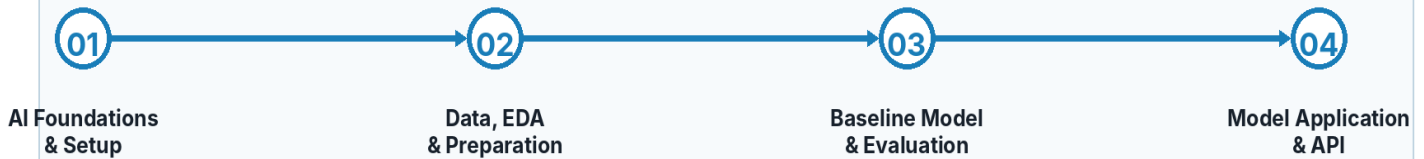
10 days • 40 hours • 2 guided projects

TRACK FOCUS

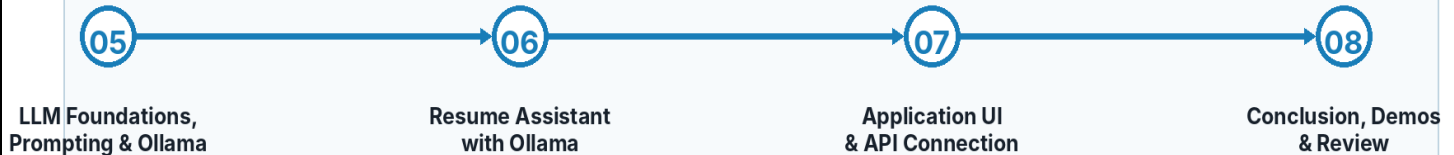
- Machine learning from raw data to a working prediction app
- LLM foundations, open-weight models and API-based application development

PROGRAMME MAP

MACHINE LEARNING PATH



LLM APPLICATION PATH



DAYS 1-2

Foundations & EDA

DAY

1

AI Foundations & Working Setup

- AI, machine learning, deep learning and generative AI
- Classification vs regression; features, targets and predictions
- Training vs inference; ML model vs LLM API
- Python, VS Code, Jupyter, Git/GitHub and Streamlit setup
- Student Performance dataset preview and problem definition

RESULT Environment ready; project workspace and dataset opened.

DAY

2

Data, EDA & Preparation

- Rows, columns, data types and the prediction target
- Missing values, duplicates and unusual values
- Simple EDA: distributions, class balance and useful relationships
- Feature selection and a clean model-ready dataset

RESULT One clean notebook with the dataset ready for model training.

DAY 1 ALSO COVERS

Programme onboarding, account checks, resource access and small setup tasks outside the technical content.

DAYS 3–4

Baseline Model & Evaluation

DAY
3

Baseline Model & Evaluation

- Train/test split and categorical encoding
- Simple preprocessing pipeline and Logistic Regression baseline
- First predictions and baseline results
- Accuracy

RESULT Baseline model trained and evaluated.

DAY
4

Model Application & API

- Save and load the trained model pipeline
- Build a simple prediction flow for new student records
- API basics: request, model input, prediction and response
- Connect the prediction flow to a simple application

RESULT ML model ready for application use.

DAYS 5–7

Large Language Models (LLMs)

DAY
5

LLM Foundations, Prompting & Ollama API

- LLMs: tokens, context, prompts and inference
- Open-weight models and where to find them: Hugging Face and Ollama
- Model size and quantization for practical local use
- Prompt structure: system/user instructions and structured output

RESULT Local LLM responding through the Ollama API.

DAY
6

AI Resume Assistant with Ollama

- Define the resume input and the feedback the assistant should return
- Send resume text to the local Ollama model
- Build a reusable prompt for strengths, gaps and improvements
- Test the assistant with different sample resumes

RESULT Resume Assistant logic working with a local LLM.

DAY
7

Application UI & API Connection

- Build a simple interface for the Resume Assistant

Review

DAY
8

Conclusion, Demos, Resources & Review

- Demo the Student Performance Prediction application
- Demo the AI Resume Assistant with the local Ollama model
- Review ML vs LLM: trained model compared with prompt + API
- Share key resources, project notes and next learning steps

RESULT Both guided projects completed and reviewed.

GUIDED PROJECTS

PROJECT 1

Student Performance Prediction System

MACHINE LEARNING

- UCI Student Performance dataset
- Data preparation, classification and evaluation
- Final model inside a web app

PROJECT 2

AI Resume Assistant

LOCAL LLM APPLICATION

- Resume text or supported document input
- Ollama local model with prompt-based feedback
- UI connected to the local API

NOTE

Student capstone projects start on Day 3 and continue alongside the guided work through the final review.

TABAARAK ICT SOLUTION
END